Andhra University College of Engineering

Computer Science and Engineering | Course Code: CS302

Database Management Systems - Relational Schema Design & Normalization 1NF

Overview: Functional dependencies, closure sets, canonical cover algorithm, lossless join decom

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: Relational Data Model & Calculus
Relational algebra operators, Tuple relational calculus, Domain relational calculus, and SQL-99 stan

Unit 2: Normalization & Functional Dependencies
Armstrong axioms, Attribute closure, 1NF, 2NF, 3NF, BCNF, and 4NF synthesis algorithms

Unit 3: Transaction Processing & Concurrency
ACID properties, Serializability, Two-Phase Locking 2PL, Timestamp ordering, and Deadlock resolution

Unit 4: Indexing & Storage Structures
B-Trees, B+ Trees indexing, Static & Dynamic Hashing, Query execution plan optimization

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Prove whether a given relational schema decomposition is lossless-join and dependency-preserving.
2. Compare strict two-phase locking Rigorous 2PL with conservative 2PL in transactional concurrency.
3. Construct a B+ Tree index with order 4 for a given set of search keys and illustrate node splits.

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* Raghu Ramakrishnan, Johannes Gehrke - Database Management Systems McGraw-Hill
* Silberschatz, Korth, Sudarshan - Database System Concepts McGraw-Hill 7th Ed

Verified NoteVault Institutional Archive - Authorized for Registered Student Use Only